



Optimizing Performance with Intersight Metrics Explorer

API Integration and Data Export

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DEVWKS-2486



Hi, nice to meet you!

My name is Marc

Passionate about **automation**

I've been working on **Compute** technologies for several years now

Find my social media:

<https://www.linkedin.com/in/marc-aeg/>

<https://github.com/mabuelgh>

slido

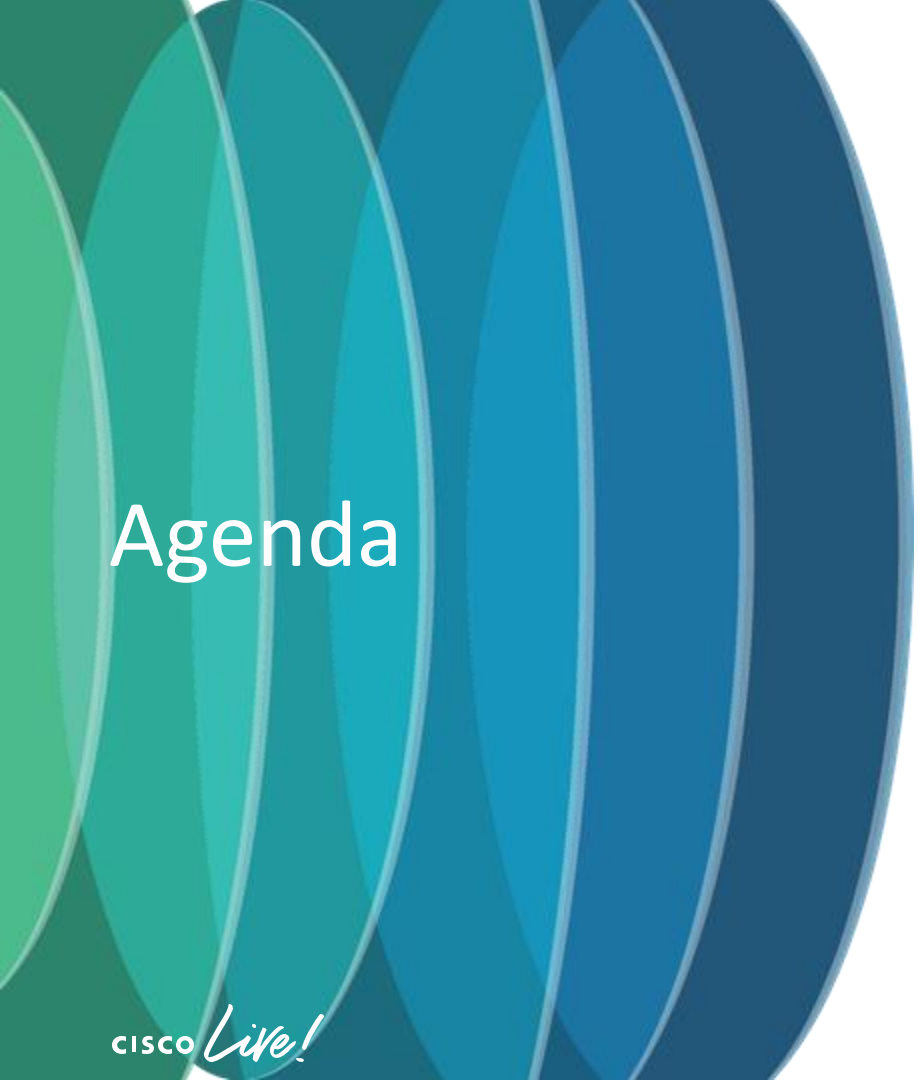
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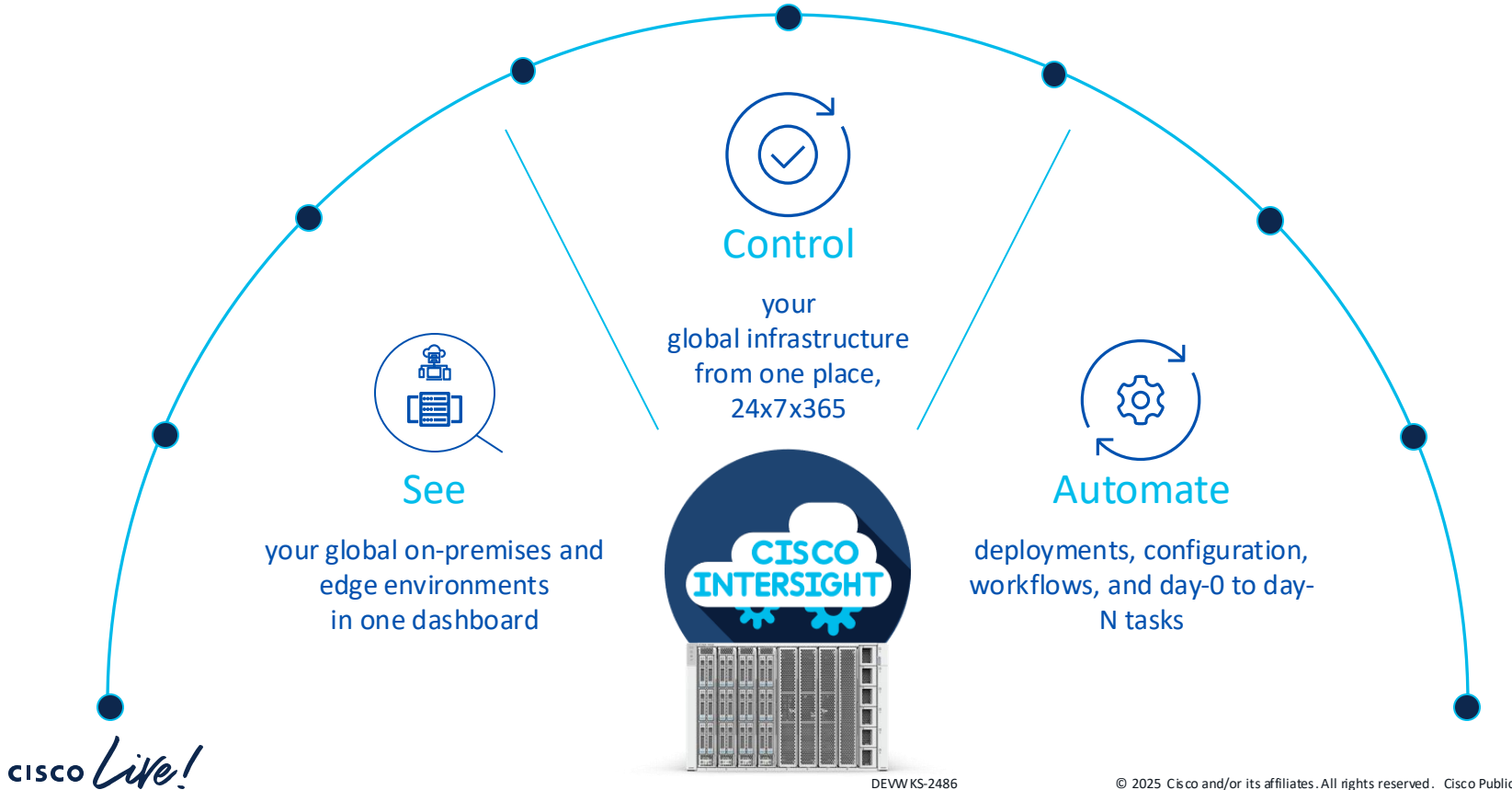
Agenda

- Intersight Data Collection Process
- Metrics Explorer
- Data Export

Intersight Data Collection Process

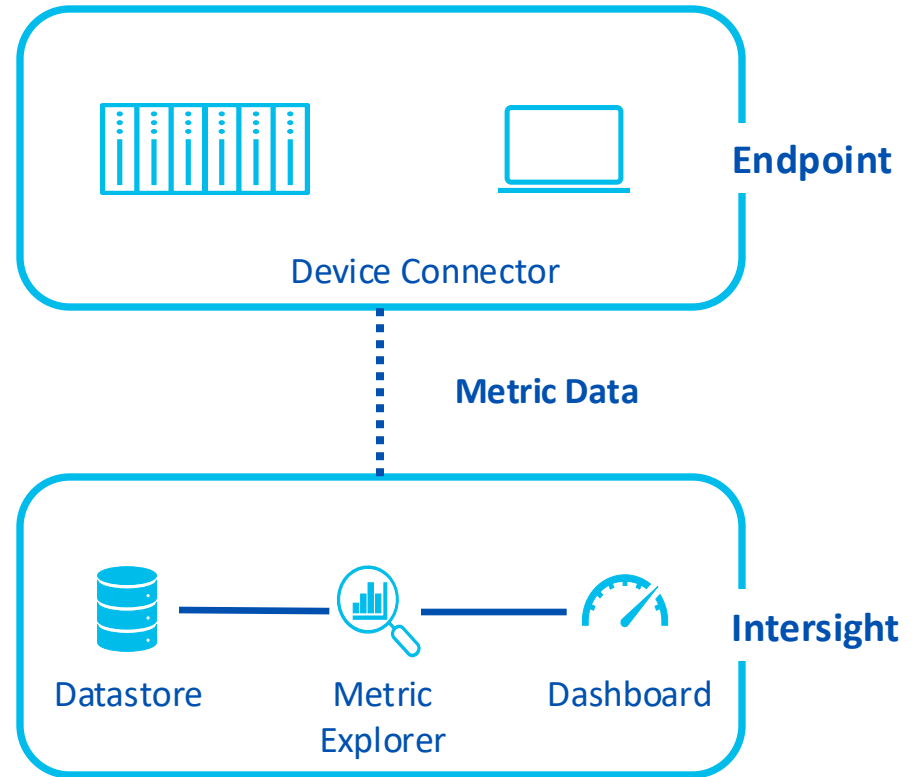


Intersight : IT Operations. Simplified.



Data Collection Process

- Provides **state**, **behavior** and **performance** metrics of devices
- **IMM** and **UMM domains** metrics are collected via the Device Connector
- Stored for a **maximum of 2 years**
- **Supports outage** with lower granularity of collection



Metrics Extracted

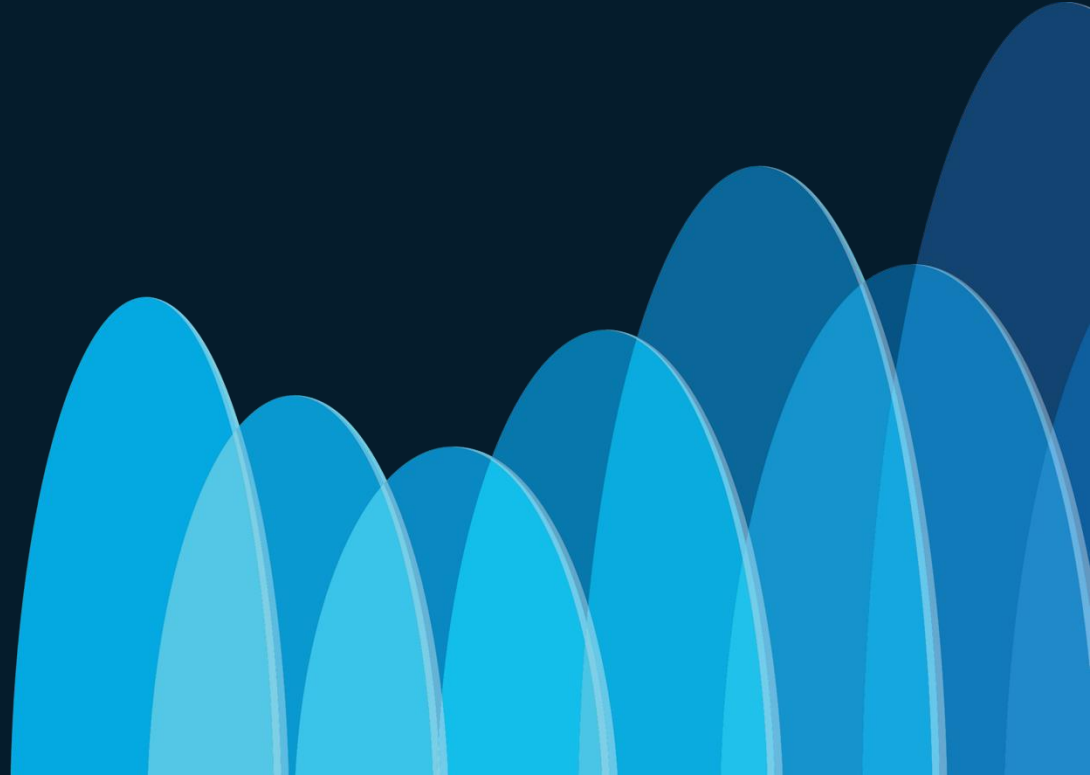
Combination of **instrument name** → **metric data** → **statistics**

- Example: Fan → Fan Speed → Minimum

Instrument Names:

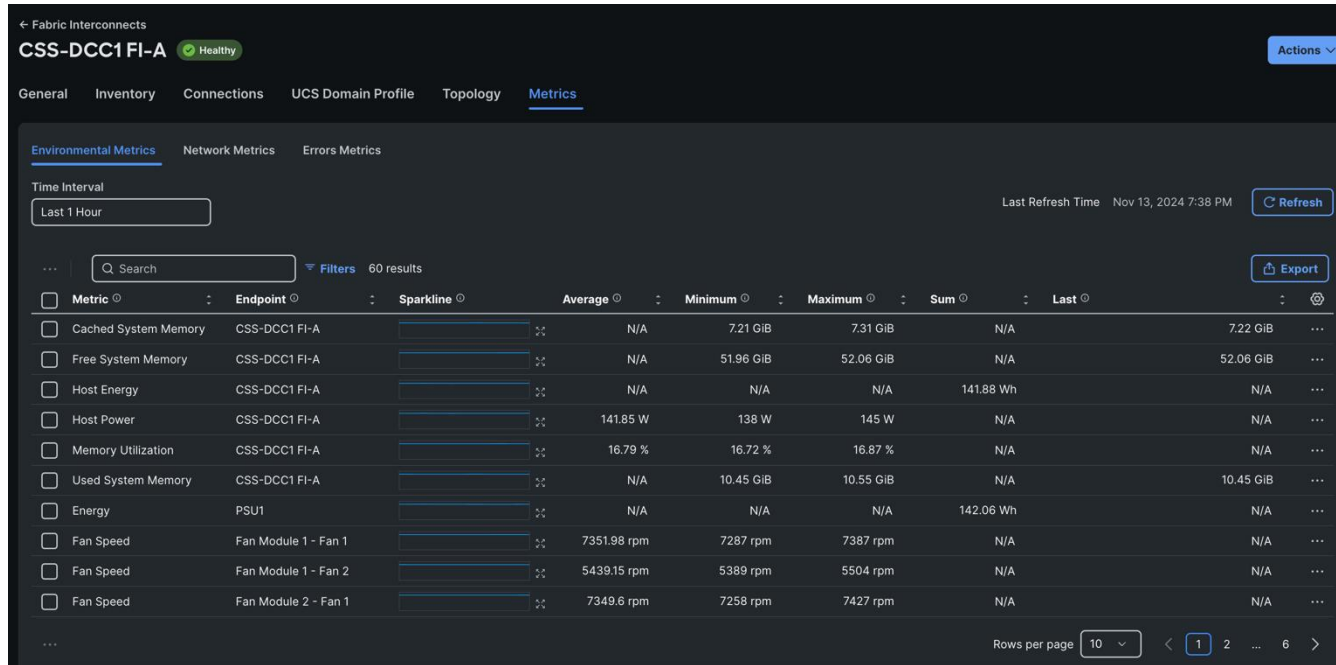
- Electric Current
- Fan
- Host Power and Status
- Memory Module
- Network Interface
- Power Supply
- Signal Power
- Temperature
- Voltage
- System CPU
- System Memory

Metrics Explorer



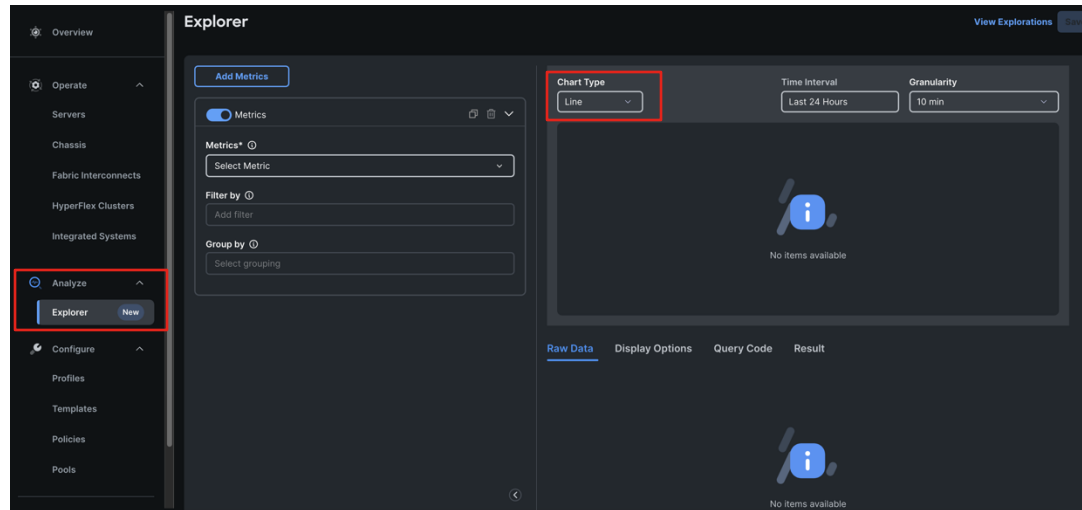
Inventory Metrics

- Metrics are available on each supported inventory object



Explorer Menu

- Metrics Explorer let you fine tune what you want to see
- Explorations can be saved for later use or as a dashboard widget



Explorer Menu – Add Metrics

- Add one or multiple metrics with filters and grouping possibilities

Add Metrics

☒ Temperature - Average

Metrics* ⓘ
Temperature - Temperature - Average

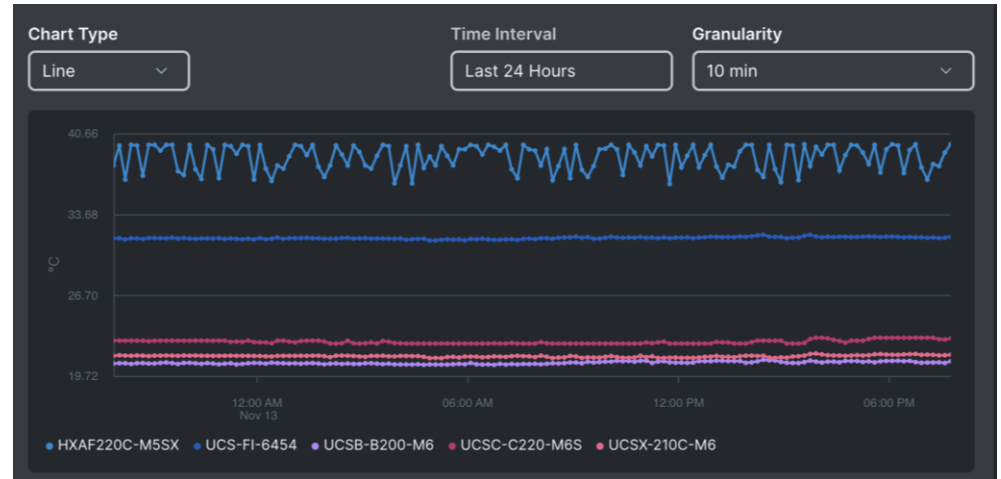
Filter by ⓘ
Add filter

Group by ⓘ
Model ×

Limit ⓘ
10

Sort by ⓘ
Descending

Aggregate Time Series ⓘ
Select Time Series Aggregation



Explorer Menu – Data view

- **Raw data** is displayed in addition to the table, can be exported as CSV
- **Original API query code** is available in addition to **JSON results**

Raw Data

Display Options

Query Code

Result

Q Search

Filters720 results

Timestamp	Label	Metric Value	Unit
Nov 12, 2024 7:58 PM	HXAF220C-M5SX	37.889	°C
Nov 12, 2024 7:58 PM	UCSF-FI-6454	31.579	°C
Nov 12, 2024 7:58 PM	UCSB-B200-M6	20.775	°C
Nov 12, 2024 7:58 PM	UCSC-C220-M6S	22.750	°C
Nov 12, 2024 7:58 PM	UCSX-210C-M6	21.450	°C

Raw Data				Display Options	Query Code	Result
1	{					
2	{					
3	"queryType": "groupBy",					
4	"dataSource": "PhysicalEntities",					
5	"granularity": {					
6	"type": "duration",					
7	"duration": 86400000,					
8	"origin": "2024-11-12T18:58:00.000Z"					
9	},					
10	"intervals": [					
11	"2024-11-12T18:58:00.000Z/2024-11-13T18:58:00.000Z"					
12	],					

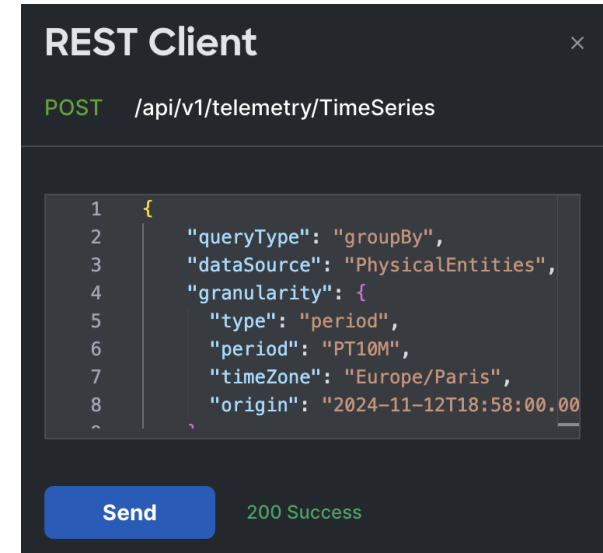
Raw Data				Display Options	Query Code	Result
1	{					
2	{					
3	"status": "fulfilled",					
4	"value": {					
5	{					
6	"version": "v1",					
7	"timestamp": "2024-11-12T19:58:00.000+01:00",					
8	"event": {					
9	"hw-temperature-Avg": 37.888888888888886,					
10	"endpoint_count": 9,					
11	"count": 9,					
12	"hw-temperature-Sum": 341,					

Data extraction



OpenTelemetry API

- Intersight's telemetry API uses **time-series data** with **OpenTelemetry standard**
- Compatible with **all OTLP** solutions
- API can be addressed with:
 - Native Druid language
 - GUI query code
 - Request on **/api/v1/telemetry/TimeSeries**



Query through REST Client

Step 1

Go to the Developer Center of Intersight to go to the REST Client
<https://intersight.com/apidocs/apirefs/>

Step 2

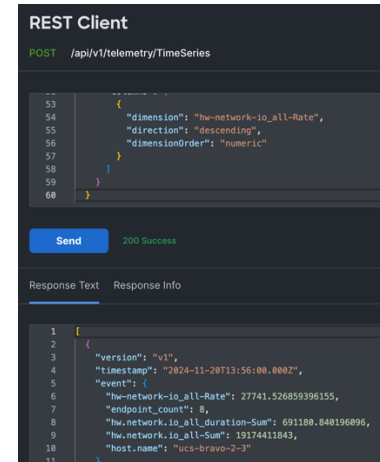
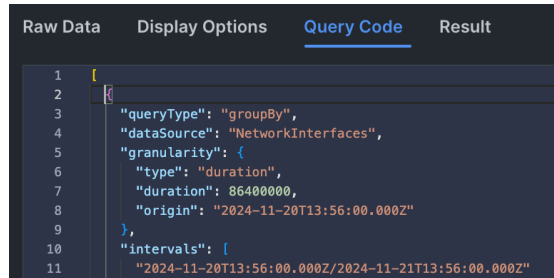
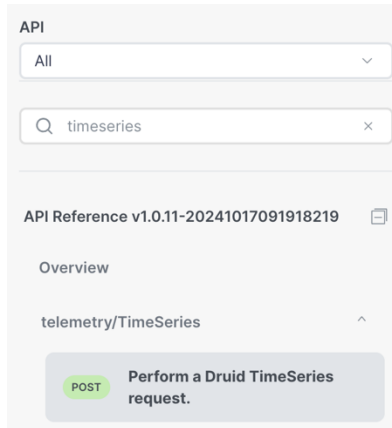
Look for the TimeSeries API Call
Click on POST method

Step 3

Copy one of the **JSON query from the Query Code** on the Explorer's GUI

Step 4

Paste the query on the right of the screen and send the POST query



Let's get bigger with InfluxDB & Grafana



Display data through Grafana – Global process



- OLTP API
- Poll metrics from claimed devices

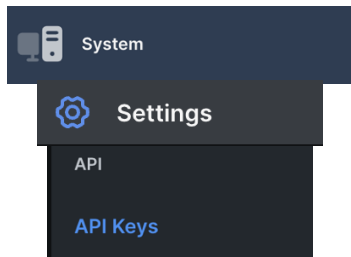
- Time Series Database
- External Python poller to populate InfluxDB

- Display data sources as charts and graphs
- Fetch data from InfluxDB

Creating an API Key

Step 1

Go to System > Settings



Step 2

Click API Keys in the left-hand navigation pane.
Click Generate API Key button on the upper-right.

Step 3

Ensure API key for OpenAPI schema version3 is selected.
Enter a Description for the key.

A screenshot of the 'Generate API Key' dialog box. It has a title bar with a close button. Below the title is an information icon and text: 'Generate API key to authenticate and authorize API requests with Intersight account.' There is a 'Description' field with a red asterisk and an info icon, containing the text 'DemoKey'. Below this is the 'API Key Purpose' section with two radio buttons: 'API key for OpenAPI schema version 3' (selected) and 'API key for OpenAPI schema version 2 (deprecated)'. There is an 'Expiration Time' field with a red asterisk and an info icon, containing the text 'Expiration Time'. At the bottom right are 'Close' and 'Generate' buttons.

Step 4

Click Generate.
Click the Save Secret Key to text file icon.

A screenshot of the 'View API Key' dialog box. It has a title bar with a close button. Below the title is an information icon and text: 'This is the only one time that the secret key can be viewed or downloaded. You cannot recover them later. However, you can create new access keys at any time.' There is an 'API Key ID' field containing the text '671235dd7564613305bf3dc7/671235de7564613305bf3dcb/6735004675646130051c7803'. Below this is a 'Secret Key' field containing the text '-----BEGIN EC PRIVATE KEY-----MIGHAgEAMBMGBYqGSM49AgEGCCqGSM49AwEHBG0wawIBAQQgA73CndnGid1AtCiLU5zujkV1Q5RrSNE4Tz0E8pKvuuhRANCAARnB5rsjq158W3q'. At the bottom is a checkbox labeled 'I have downloaded the Secret Key.'.

Display data through Grafana

Step 1

Download project from
<https://github.com/mabuelg/intersight-metrics-bridge>

Step 2

Launch initial setup script
Add intersight API Keys and list of
servers in inventory yaml file

Step 3

Use docker to compose the 3
containers

Step 4

Make sure the containers are
up and running
Access Grafana on
127.0.0.1:3000



influxdb

RUNNING PORT 8086



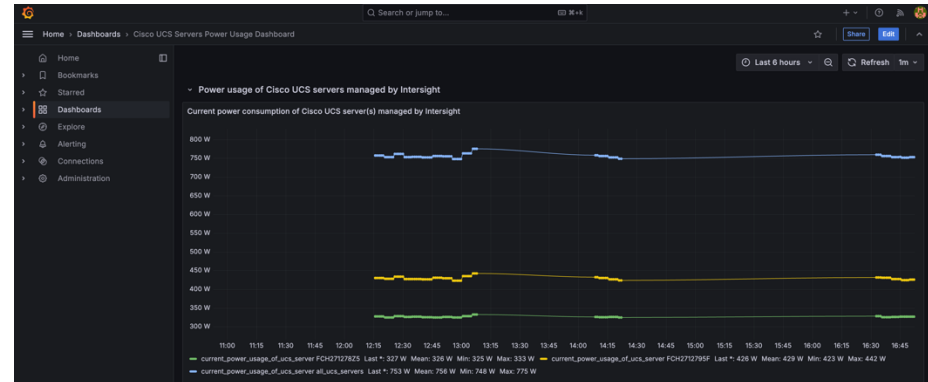
grafana

RUNNING PORT 3000



intersight-poller

RUNNING PORT 5001



Next Steps

You have the **cooking recipe**, now let's do it yourself at home:

- **Enable** metric collection (if not)
- **Try** the metric explorer with your data
- **Use** the API to collect and send data to your observability platform

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- Visit the On-Demand Library for more sessions at ciscolive.com/on-demand. Sessions from this event will be available from March 3.

Contact me at: mabuelgh@cisco.com

Webex App

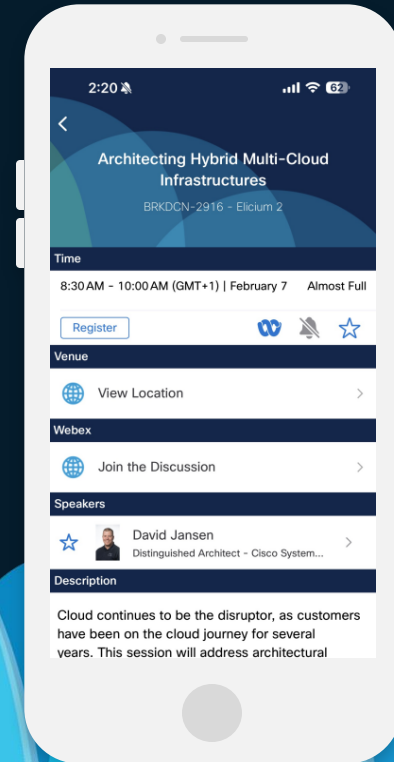
Questions?

Use the Webex app to chat with the speaker after the session

How

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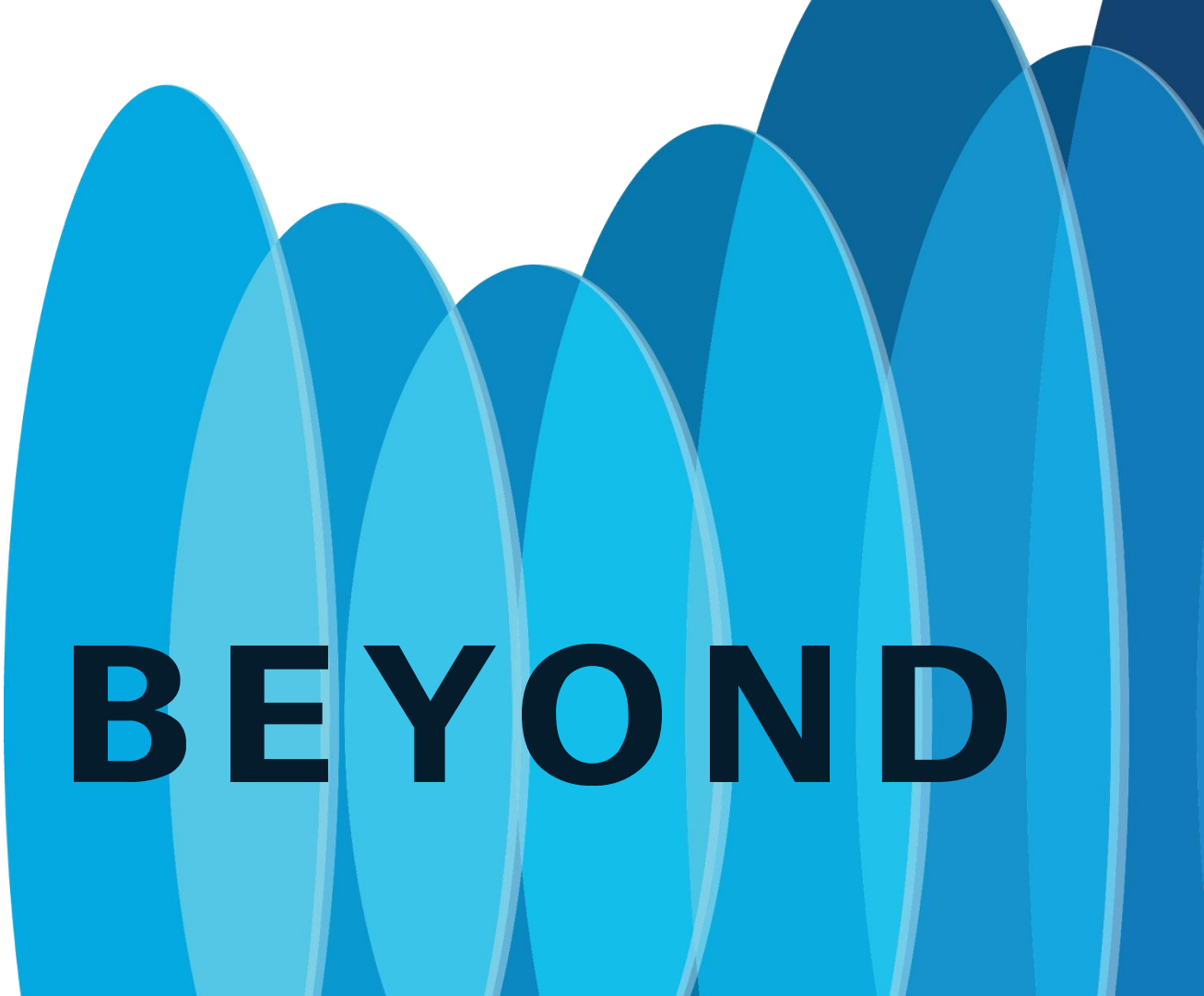




Thank you

CISCO *Live!*

GO BEYOND

The background of the slide features a series of overlapping, teardrop-shaped elements in various shades of blue, ranging from a light sky blue to a deep navy blue. These shapes are arranged in a way that creates a sense of depth and movement, resembling a stylized horizon or a series of waves. The overall composition is clean and modern, with the text elements clearly legible against the white background.